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CUSTOMER : IDAMAN PHARMA MANUFACTURING SDN BHD
DESCRIPTION : Phenobarbitone Tablet 30mg
MEASUREMENT : W: 148MM X H: 210MM
MATERIAL : SMPO 050
COLOUR : BLACK
FINISHING : -
MATERIAL CODE:
Lt-105.06
Attn : MS ZAKIAH
From : MALA
I hereby verify & confirm that this draft is the final copy. Any subsequence changes of the design and content of the end product will not be borne by THUNDER PRINT SDN. BHD.
Customer Signature & Date
21.2.2023

pharmaniaga®

IDAMAN PHARMA PHENOBARBITONE TABLET 30MG

DESCRIPTION
A clean, dry, round, uniform, biconvex, even-edged tablets and white in colour.

COMPOSITION
Each tablet contains:
- Active Ingredient : Phenobarbitone 30mg
- Preservatives : Sodium Benzoate 0.03mg

PHARMACODYNAMICS
Phenobarbitone is a long-acting barbiturate. It is widely used in the treatment of epilepsy due to its depressant effect on motor cortex. It also has sedative effect and some protective action against all varieties of human partial and generalised epilepsy. Phenobarbitone proven to have inconsistent effect on suppressing experimental epileptic foci and epileptic after discharges however, it inhibits synaptic transmission at spinal cord. The drug acts by prolonging opening of Cl- ion channels in postsynaptic neuronal membranes which leads to hyperpolarisation and impairs nerve impulse propagation. Phenobarbitone also decreases inter-neuronal Na+ concentrations and inhibits Ca+ influx into depolarised synaptosomes. The drug increases serotonin level and inhibits noradrenaline (norepinephrine) reuptake into synaptosomes.

PHARMACOKINETICS
Absorption
Phenobarbitone is readily absorbed from gastro-intestinal tract. It is relatively lipid-insoluble. Peak concentrations are achieved in about 2 hours after oral dose administration.
Distribution
Phenobarbitone is about 45-60% bound to plasma proteins. It crosses the placenta and is present in breast milk. Phenobarbitone is distributed to all tissues and fluid with high concentration in the brain, liver and kidneys.
Metabolism
Plasma half-life of phenobarbitone is about 75 to 120 hours in adults and about 21-75 hours in children. It is greatly prolonged in neonates. Phenobarbitone kinetics show considerable interindividual variation. It is only partly metabolised in the liver.
Elimination
Approximately 25% of the phenobarbitone dose is excreted unchanged in the urine at normal urinary pH.

INDICATIONS
Insomnia: For short term treatment of insomnia.
Sedation: To relieve anxiety, tension and apprehension.
Seizures: Indicated as a long term anticonvulsants therapy for the treatment of generalised tonic clonic and simple partial (cortical focal) seizures. It can also be used in the prophylaxis and treatment of febrile seizures.
Hyperbilirubinemia: Used in the prevention and treatment of hyperbilirubinemia in neonates. It is being used also to lower Bilirubin concentrations in patients with congenital non-haemolytic unconjugated hyperbilirubinemia or chronic intrahepatic cholestasis.

RECOMMENDED DOSAGE
Usual Dose: 30 to 125mg, 3 times daily.
Up to 350mg daily can be given, in divided doses.

ROUTE OF ADMINISTRATION
Oral

CONTRAINDICATIONS
Phenobarbitone is contraindicated in patients:
- hypersensitivity to phenobarbitone or any barbiturates
- Porphyria, acute intermittent or history of this disease
- Severe respiratory depression
- Severe renal or hepatic impairment

WARNINGS AND PRECAUTIONS
Suicidal Ideation and Behaviour
Suicidal ideation and behaviour have been reported in patients treated with anti-epileptic agents in several indications. A meta-analysis of randomised placebo-controlled trials of anti-epileptic drugs has also shown a small increased risk of suicidal ideation and behaviour. The mechanism of this risk is not known and the available data do not exclude the possibility of an increased risk for phenobarbitone.

Therefore, patients should be monitored for signs of suicidal ideation and behaviours and appropriate treatment should be considered. Patients and caregivers of patients should be advised to seek medical advice should signs of suicidal ideation or behaviour emerge.

Steven-Johnson Syndrome (SJS) and Toxic Epidermal Necrolysis (TEN)
Life-threatening cutaneous reactions Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN) have been reported with the use of phenobarbitone. Patients should be advised of the signs and symptoms and monitored closely for skin reactions. The highest risk for occurrence of SJS or TEN is within the first weeks of treatment.

If symptoms or signs of SJS or TEN (e.g., progressive skin rash often with blisters or mucosal lesions) are present, phenobarbitone treatment should be discontinued. The best results in managing SJS and TEN come from early diagnosis and immediate discontinuation of any suspect drug. Early withdrawal is associated with a better prognosis.

If the patient has developed SJS or TEN with the use of phenobarbitone, phenobarbitone must not be re-started in this patient at any time.

Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS)
Multi-organ hypersensitivity reactions, also known as drug reaction with eosinophilia and systemic symptoms (DRESS), have occurred with the use of phenobarbitone. Some have been fatal or life-threatening. DRESS typically, although not exclusively, presents with fever, rash, and/or lymphadenopathy in association with other organ system involvement, such as hepatitis, nephritis, hematologic abnormalities, aseptic meningitis, myocarditis, or myositis, sometimes resembling an acute viral infection. Eosinophilia is often present. This disorder is variable in its expression and other organ systems not noted here may be involved. The syndrome shows wide spectrum of clinical severity and may rarely lead to disseminated intravascular coagulation and multi organ failure.

It is important to note that early manifestation of hypersensitivity (e.g fever, lymphadenopathy) may be present even though rash is nor evident. If such signs or symptoms present, the patient should be evaluated immediately. Phenobarbitone should be discontinued.

Bone Disorder
Long-term use of antiepileptic such as carbamazepine, phenobarbitone, phenytoin, primidone, oxcarbazepine, lamotrigine, and sodium valproate are associated with a risk of decreased bone mineral density that may lead to weakened or brittle bones. Discontinuation of phenobarbitone should be considered if evidence of significant bone marrow depression develops.

Care should be used in the following situations:
- Patients with the rare hereditary problems of galactose intolerance, the lapp lactase deficiency or glucose – galactose malabsorption should not take this medicine;
- Respiratory depression (avoid if severe);
- Young, debilitated or senile patients;
- Renal impairment;
- Existing liver disease;
- Sudden withdrawal should be avoided as severe withdrawal syndrome (rebound insomnia, anxiety, tremor, dizziness, nausea, fits and delirium) may be precipitated;
- Acute chronic pain – paradoxical excitement may be induced or important symptoms masked;
- Prolonged use may result in dependence of the alcohol-barbiturate type. Care should be taken in treating patients with a history of drug abuse or alcoholism.

Labour and Delivery
Full anaesthetic doses of barbiturates decrease the force and frequency of uterine contractions - use of barbiturates during labour may cause respiratory depression in the neonate, especially the premature neonate, because of immature hepatic function.

Paediatrics
Some children may react to barbiturates with paradoxical excitement.

Geriatrics
Geriatric patients may be more sensitive to the effects of barbiturates. They may react to usual doses of barbiturates with excitement, confusion or depression.

INTERACTIONS WITH OTHER MEDICAMENTS
Alcohol
Concurrent administration with alcohol may lead to an additive Central Nervous System (CNS) depressant effect. This is likely with concurrent administration with other CNS depressants.

Antidepressants
Including Monoamine Oxidase Inhibitors (MAOIs), Selective Serotonin Reuptake Inhibitor (SSRIs) and tricyclics may antagonize the antiepileptic activity of phenobarbitone by lowering the convulsive threshold

Antiepileptics
Phenobarbitone plasma concentrations increased by oxcarbazepine, phenytoin and sodium valproate. Vigabatrin possibly decreases phenobarbitone plasma concentrations.

Antipsychotics
Concurrent use of chlorpromazine and thioridazine with phenobarbitone can reduce the serum levels of either drug.

Folic acid
If folic acid supplements are given to treat folate deficiency, which can be caused by the use of phenobarbitone, the serum phenobarbitone levels may fall, leading to decreased seizure control in some patients.

Memantine
The effect of phenobarbitone is possibly reduced.

Methylphenidate
Plasma concentration of Phenobarbitone is possibly increased.

Anti-arrhythmics
Disopyramide and quinidine loss of arrhythmia control is possible. Plasma levels of antiarrhymics should be monitored, if phenobarbitone is added or withdrawn. Changes in dosage may be necessary.

Antibacterials
Chloramphenicol, doxycycline, metronidazole and rifampicin. Avoid concomitant use of telithromycin during and for 2 weeks after Phenobarbitone.

Anticoagulants.
Phenobarbitone increases the rate of metabolism reducing serum concentrations when taking together with anticoagulants.

Antidepressants
Phenobarbitone increases the rate of metabolism reducing serum concentrations when taking together with paroxetine, mianserin and tricyclic antidepressants.

Antiepileptics
Phenobarbitone increases the rate of metabolism reducing serum concentrations when taking together with carbamazepine, lamotrigine, tiagabine, zonisamide, primidone and possibly ethosuxamide.

Antifungals
Antifungal effects of griseofulvin can be reduced or even abolished by concurrent use. Phenobarbitone possibly reduces plasma concentrations of itraconazole or posaconazole. Avoid concomitant use of voriconazole.

Antipsychotics
Phenobarbitone possibly reduces concentration of aripiprazole

Antivirals
Phenobarbitone possibly reduces plasma levels of abacavir, amprenavir, darunavir, lopinavir, indinavir, nelfinavir, saquinavir.

Anxiolytics and Hypnotics
Phenobarbitone increases the rate of metabolism reducing serum concentrations when taking together with clonazepam.

Aprepitant
Phenobarbitone possibly reduces plasma concentration of aprepitant.

Beta-blockers
Phenobarbitone increases the rate of metabolism reducing serum concentrations when taking together with metoprolol, timolol and possibly propranolol.

Calcium channel blockers
Phenobarbitone causes reduced levels of felodipine, isradipine, diltiazem, verapamil, nimodipine and nifedipine and an increase in dosage may be required.

Cardiac Glycosides
Blood levels of digitoxin can be halved by concurrent use.

Ciclosporin or tacrolimus.
Phenobarbitone increases the rate of metabolism reducing serum concentrations when taking together with ciclosporin or tacrolimus

Corticosteroids
Phenobarbitone increases the rate of metabolism reducing serum concentrations when taking together with corticosteroids.

Cytotoxics
Phenobarbitone possibly reduces the plasma levels of etoposide or irinotecan.

Diuretics
Concomitant use with eplerenone should be avoided.

Haloperidol
Serum levels are approximately halved by concurrent used with phenobarbitone.

Hormone Antagonists
Phenobarbitone increases the rate of metabolism reducing serum concentrations when taking together with gestrinone and possibly toremifene.

Methadone
Levels can be reduced by concurrent use of phenobarbitone and withdrawal symptoms have been reported in patients maintained on methadone when phenobarbitone has been added. Increases in the methadone dosage may be necessary.

Montelukast
Phenobarbitone increases the rate of metabolism reducing serum concentrations when taking together with montelukast.

Oestrogens and Progestogens
Reduced contraceptive effect.

Sodium oxybate
Enhanced effects, avoid concomitant use.

Theophylline
May require an increase in theophylline dose.

Thyroid hormones
May increase requirements for thyroid hormones in hypothyroidism.

Tibolone
Phenobarbitone increases the rate of metabolism reducing serum concentrations when taking together with tibolone

Tropisetron
Phenobarbitone increases the rate of metabolism reducing serum concentrations when taking together with tropisetron.

Vitamins
Barbiturates possibly increase requirements for vitamin D.

PREGNANCY AND LACTATION

Pregnancy
Phenobarbitone therapy in epileptic pregnant women presents a risk to the foetus in terms of major and minor congenital defects such as congenital craniofacial, digital abnormalities and, less commonly, cleft lip and palate. The risk of teratogenic effects developing appears to be greater if more than one antiepileptic drug is administered. The risk to the mother, however is greater if phenobarbitone is withheld and seizure control is lost. The risk-benefit balance, in this case, favours continued use of the drug during pregnancy at the lowest possible level to control seizures.

Patients taking phenobarbitone should be adequately supplemented with folic acid before conception and during pregnancy. Folic acid supplementation during pregnancy can help to reduce the risk of neural defects to the infant.

Phenobarbitone readily crosses the placenta following oral administration and is distributed throughout foetal tissue, the highest concentrations being found in the placenta, foetal liver and brain. Adverse effects on neurobehavioral development have also been reported.

Haemorrhage at birth and addiction are also a risk. Prophylactic treatment with vitamin K^1 for the mother before delivery (as well as the neonate) is recommended, the neonate should be monitored for signs of bleeding.

Lactation
Phenobarbitone is excreted into breast milk and there is a small risk of neonatal sedation. Phenobarbitone is eliminated slowly in neonates and may accumulate. Therefore, the benefits of breast feeding should be weighed against the possible risks to the infant and a decision should be made whether to discontinue nursing or to discontinue phenobarbitone, taking into account the importance of the drug to the mother. Breastfed infants should be observed for excessive drowsiness, dizziness, feeding problems, allergic skin reactions such as rash or other adverse reactions. If any of these occur, breastfeeding should be discontinued. When breastfeeding is discontinued, there is a potential for withdrawal symptoms in infants.

SIDE EFFECTS / ADVERSE REACTIONS

Blood and the lymphatic system disorders:
Megaloblastic anaemia (due to folate deficiency), agranulocytosis, thrombocytopenia.

Musculoskeletal and connective tissue disorders
Dupuytren's contracture, frozen shoulder, arthralgia, osteomalacia, rickets.

There have been reports of decreased bone mineral density, osteopenia, osteoporosis and fractures in patients on long-term therapy with phenobarbitone. The mechanism by which phenobarbitone affects bone metabolism has not been identified.

Reproductive and breast disorders
Peyronie's disease.

Psychiatric disorder
Paradoxical reaction (unusual excitement), hallucinations, restlessness and confusion in the elderly, mental depression, memory and cognitive impairment, drowsiness, lethargy, feeling faint, headache, nightmare or trouble sleeping and unusual irritability.

Nervous system disorders
Hyperactivity, behavioural disturbances in children, ataxia, nystagmus.

Cardiac disorders
Hypotension.

Respiratory disorders
Respiratory depression.

Hepato-biliary
Hepatitis, cholestasis.

Skin and subcutaneous tissue disorders
Allergic skin reactions (maculopapular morbilliform or scarlatiniform rashes), other skin reactions such as exfoliative dermatitis, erythema multiforme.

Severe cutaneous adverse reactions (SCARs): Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN) have been reported.

General disorders and administration site conditions
Antiepileptic hypersensitivity syndrome (features include fever, rash, lymphadenopathy, lymphocytosis, eosinophilia, haematological abnormalities, hepatic and other organ involvement including renal and pulmonary systems which may become life threatening), nausea, vomiting.

SYMPTOMS AND TREATMENT OF OVERDOSE

Toxicity varies between patients; tolerance will develop with chronic use. Features of poisoning are to be expected after ingestion of 1g in adults.

Symptoms
Drowsiness, dysarthria, ataxia, nystagmus and disinhibition. There may also be coma, cardiovascular collapse, cardiac arrest, hypotension, hypotonia, hyporeflexia, rhabdomyolysis, hypothermia, hypotension and respiratory depression.

Barbiturates decrease gut motility, which may lead to slow onset and worsening of symptoms or cyclical improvement and worsening of symptoms.

Treatment
Treatment of barbiturate overdosage is primarily supportive. An adequate airway should be maintained, with assisted respiration and administration of oxygen as needed. Vital signs and fluid balance should be monitored. If the patient is conscious and has not lost the gag reflex, emesis may be induced with ipecac syrup, care should be taken to prevent pulmonary aspiration of vomitus.

After vomiting is completed, activated charcoal in a glass of water or sorbitol may be administered to prevent absorption and increase excretion of the barbiturate. Activated charcoal (50g for an adult, 10-15g for kids under 5 years) can be considered if >10mg/kg body weight of drug has been ingested within 1 hour, provided airway can be protected. Repeat dose activated charcoal is the best method of enhancing elimination of phenobarbitone in symptomatic patients.

If emesis is contraindicated, gastric lavage may be performed with a cuffed endotracheal tube in place with the patient face down. Activated charcoal should be left in the stomach and a saline cathartic may be administered.

Charcoal haemoperfusion is the treatment of choice for the majority of patients with severe barbiturate poisoning who fail to improve, or who deteriorate despite good supportive care.

Fluid therapy and other standard treatment of shock should be administered, if necessary.

A vasopressor may be required if hypotension occurs. If renal function is normal, forced diuresis may help to eliminate the barbiturate. Alkalinization of the urine increases renal excretion of some barbiturates, especially phenobarbitone. Treat rhabdomyolysis with urinary alkalinisation.

Fluid or sodium overload should be avoided, especially if cardiovascular status is decreased although hemodialysis or hemoperfusion is not recommended as a routine procedure, it may be used in severe barbiturate poisoning or if the patient is anuric or in shock or in cases of acute renal or severe hyperkalaemia. Chest physiotherapy should be administered.

In severe hypotension dopamine or dobutamine can be used.

If pneumonia is suspected, appropriate care should be taken to prevent hypostatic pneumonia, decubiti, aspiration and other complications that may occur with altered states of consciousness.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Phenobarbitone may impair the mental and/or physical abilities required for the performance of potentially hazardous tasks such as driving a car or operating machinery. Patients should be advised to make sure they are not affected before undertaking any potentially hazardous tasks.

STORAGE CONDITIONS

Keep in a dry place, below 30ºC; Protect from light.

PACK SIZE

Blister Pack of 10's (5 Strips of 10 Tablets)

SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

REGISTRATION NUMBER

MAL19860219AZ

CONTROLLED MEDICINE
UBAT TERKAWAL

KEEP MEDICINES OUT OF REACH OF CHILDREN
JAUHI UBAT-UBATAN DARI KANAK-KANAK

For further information, please consult your doctor or your pharmacist.

Revision Number : 08
Revision Date : 21 Feb 2023

Lt-105.06 PRODUCT REGISTRATION HOLDER AND MANUFACTURER

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